Practical No. 1

Title

Present Historical Development of Contemporary Research and Identify Evidences of Research in Ayurveda Classical Literature (with relevance to Computer Applications)

Aim

To study the historical development of contemporary research and identify evidence of research in Ayurveda classical literature while understanding its relevance to research methodologies used in Computer Applications.

Objectives

1. To understand the evolution of research from ancient times to the modern era.
2. To identify research practices mentioned in Ayurveda classical literature.
3. To compare traditional research methods with modern scientific research.
4. To understand the importance of systematic research in Computer Applications.
5. To develop research and documentation skills using digital resources.

Requirements

Hardware

- Computer/Laptop
- Internet Connection

Software

- Google Chrome / Microsoft Edge
- Microsoft Word / Google Docs
- Microsoft PowerPoint (Optional)

Resources

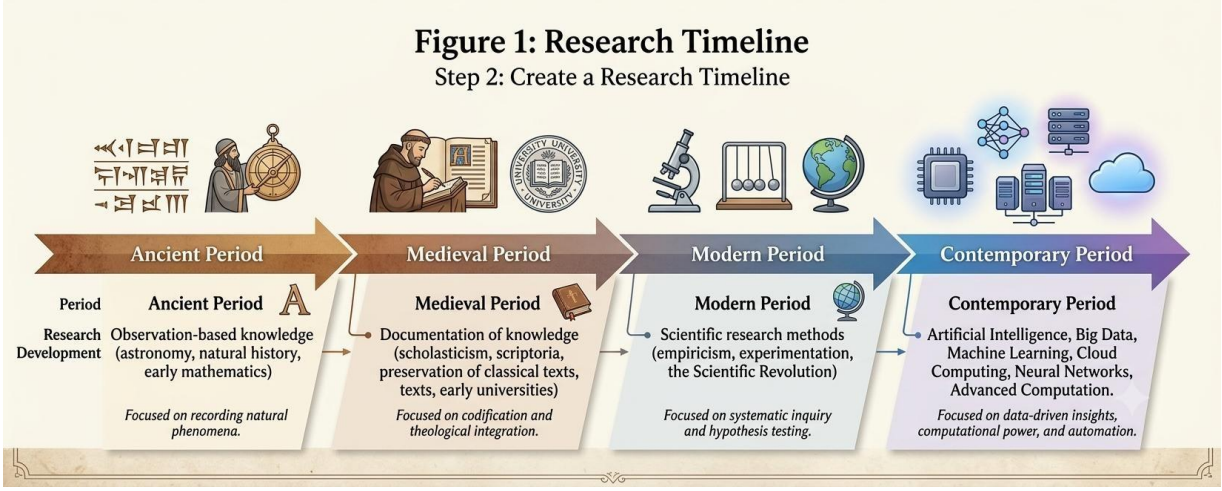
- Google Scholar
- Research Journals
- Ayurveda Classical Literature (Charaka Samhita, Sushruta Samhita, Ashtanga Hridaya)
- Research Methodology Textbook

Practical Activity

Step 1: Study Contemporary Research

- Study the meaning and importance of contemporary research.
- Understand how research has evolved from ancient observations to modern scientific methods.

Step 2: Create a Research Timeline



Step 3: Study Ayurveda Classical Literature

Study the following classical books:

- Charaka Samhita
- Sushruta Samhita
- Ashtanga Hridaya

Identify examples of:

- Observation
- Experimentation
- Diagnosis
- Documentation
- Evidence-based treatment

Step 4: Compare Ancient and Modern Research

Ayurveda Research	Computer Application Research
Observation of symptoms	Data Collection
Diagnosis	Data Analysis
Treatment	Software Solution
Documentation	Research Paper
Validation	Testing and Evaluation

Step 5: Prepare a Short Report

Title:

Present Historical Development of Contemporary Research and Identify Evidences of Research in Ayurveda Classical Literature

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Introduction

Research is a systematic process of collecting, analyzing, and interpreting information to discover new knowledge or solve problems. It helps in improving existing knowledge and developing innovative solutions. In the modern world, research plays an essential role in science, medicine, engineering, and computer applications.

The concept of research is not new. Ancient civilizations, especially India, practiced systematic observation and experimentation in various fields. Ayurveda, one of the oldest medical systems, contains evidence of research through careful observation of diseases, diagnosis, treatment methods, and documentation of results. Contemporary research follows scientific methods such as data collection, experimentation, analysis, validation, and reporting, which are also reflected in ancient Ayurvedic practices.

Historical Development of Research

The development of research has taken place over several centuries.

1. Ancient Period

Research began with observation of nature and human life. Ancient scholars recorded their experiences and developed knowledge through observation, experimentation, and logical

reasoning. Ayurveda texts such as **Charaka Samhita** and **Sushruta Samhita** are examples of early research documentation.

2. Medieval Period

During this period, knowledge was preserved and expanded through books and manuscripts. Researchers focused on documenting existing knowledge, improving medical practices, and sharing findings among scholars.

3. Modern Period

Research became more scientific with the introduction of the scientific method. Experiments, hypothesis testing, statistical analysis, and laboratory investigations became essential components of research.

4. Contemporary Period

Today, research is technology-driven. Advanced tools such as Artificial Intelligence (AI), Machine Learning (ML), Big Data, Cloud Computing, and Data Analytics are widely used. Researchers collaborate globally, publish research papers online, and use digital databases for literature review and data analysis.

Evidence of Research in Ayurveda Classical Literature

Ayurveda provides several examples of systematic research practices.

- **Observation:** Ancient physicians carefully observed patients' symptoms before diagnosing diseases.
- **Experimentation:** Different herbal medicines and treatment methods were tested to determine their effectiveness.
- **Documentation:** Medical knowledge was recorded in classical texts such as **Charaka Samhita**, **Sushruta Samhita**, and **Ashtanga Hridaya**.
- **Diagnosis:** Diseases were classified based on symptoms, causes, and patient conditions.
- **Validation:** Treatments were continuously evaluated based on patient recovery and clinical observations.

These practices demonstrate that Ayurveda followed a structured research methodology similar to modern scientific research.

Comparison with Computer Application Research

Although Ayurveda and Computer Applications belong to different fields, both follow a systematic research process.

Ayurveda Research	Computer Application Research
Observation of symptoms	Data collection
Disease diagnosis	Problem identification
Selection of treatment	Algorithm or solution design
Testing medicines	Software testing
Recording treatment results	Documentation and reporting
Validation through patient recovery	Performance evaluation and validation

In Computer Applications, researchers collect data, develop algorithms, test software, analyze results, and document their findings. Similarly, Ayurveda researchers observed patients, selected treatments, evaluated outcomes, and documented their knowledge. Both approaches rely on evidence-based decision-making and continuous improvement.

Conclusion

The study of historical development shows that research has evolved from simple observations in ancient times to highly advanced scientific investigations supported by modern technology. Ayurveda classical literature demonstrates that systematic observation, experimentation, documentation, diagnosis, and validation were practiced thousands of years ago, making it one of the earliest examples of organized research.

For Computer Application students, understanding the evolution of research helps in developing strong research skills, improving problem-solving abilities, and conducting quality projects. Modern research in computer science follows the same fundamental principles of observation, data collection, analysis, testing, validation, and documentation that were practiced in ancient research systems. Therefore, both traditional and contemporary research emphasize systematic investigation to generate reliable knowledge and solve real-world problems.

Expected Output

After completing the practical, the student should be able to:

- Explain the historical development of research.
- Identify research evidence in Ayurveda classical literature.
- Compare ancient research methods with modern research methodologies.
- Understand how research methodology is applied in Computer Applications.
- Prepare a structured research report using digital tools.

Result

The historical development of contemporary research was successfully studied. Evidence of systematic research practices in Ayurveda classical literature was identified and compared with modern research methodologies used in Computer Applications. The practical demonstrated that both traditional and modern research follow systematic processes involving observation, data collection, analysis, validation, and documentation, although modern research extensively utilizes digital technologies and computational tools.

Practical No. 2

Title:-

Illustrate Primary and Secondary Research and Differentiate Between Them with Suitable Examples

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Aim

To understand the concepts of **Primary Research and Secondary Research**, study their methods and applications, and differentiate between them with suitable examples.

Objectives

1. To understand the meaning and purpose of primary research.
2. To understand the meaning and purpose of secondary research.
3. To identify different methods used for collecting primary and secondary data.
4. To differentiate between primary and secondary research.
5. To understand the advantages and limitations of both research methods.
6. To identify suitable research methods for computer application-related studies.

Requirements

Hardware

- Computer/Laptop
- Internet connection

Software/Tools

- MS Word / Google Docs
- MS Excel / Google Sheets
- Web browser
- MS PowerPoint

Reference Sources

- Research methodology textbooks
- Research papers and journals
- Government reports
- Academic websites and databases
- Books and reliable online sources

Introduction

Research requires the collection of information to answer a research question or solve a particular problem. Based on the source of information, research data can broadly be collected through **Primary Research** and **Secondary Research**.

Primary Research involves collecting new and original information directly from respondents, experiments, observations, surveys, interviews, or other first-hand sources.

Secondary Research involves using information that has already been collected, analyzed, or published by other researchers, organizations, institutions, or government agencies.

Both methods are important in research. A researcher may use either method independently or combine both methods depending on the research problem.

Primary Research

Definition

Primary Research is the process of collecting **original and first-hand data directly from the source** for a specific research purpose.

The researcher personally collects the information according to the requirements of the study.

Common Methods of Primary Research

1. **Questionnaires/Surveys**
2. **Interviews**
3. **Observation**

4. **Experiments**
5. **Focus Groups**
6. **Field Studies**

Example

Suppose an M.Sc. Computer Applications student wants to study:

"Student Satisfaction with Online Learning Platforms."

The researcher can create a Google Form and collect responses directly from 100 students.

The collected responses are **primary data** because the researcher obtained them directly from the participants for the current study.

Secondary Research

Definition

Secondary Research is the process of using **existing information or data that has already been collected and published by someone else**.

The researcher does not directly collect the original information from respondents.

Sources of Secondary Research

1. Research papers
2. Books
3. Government reports
4. Websites
5. Journals
6. Company reports
7. Previous research studies
8. Statistical databases
9. Newspapers and magazines
10. Institutional reports

Example

For the same study on:

"Student Satisfaction with Online Learning Platforms."

A researcher may use previously published research papers, government education reports, or statistical reports about online education.

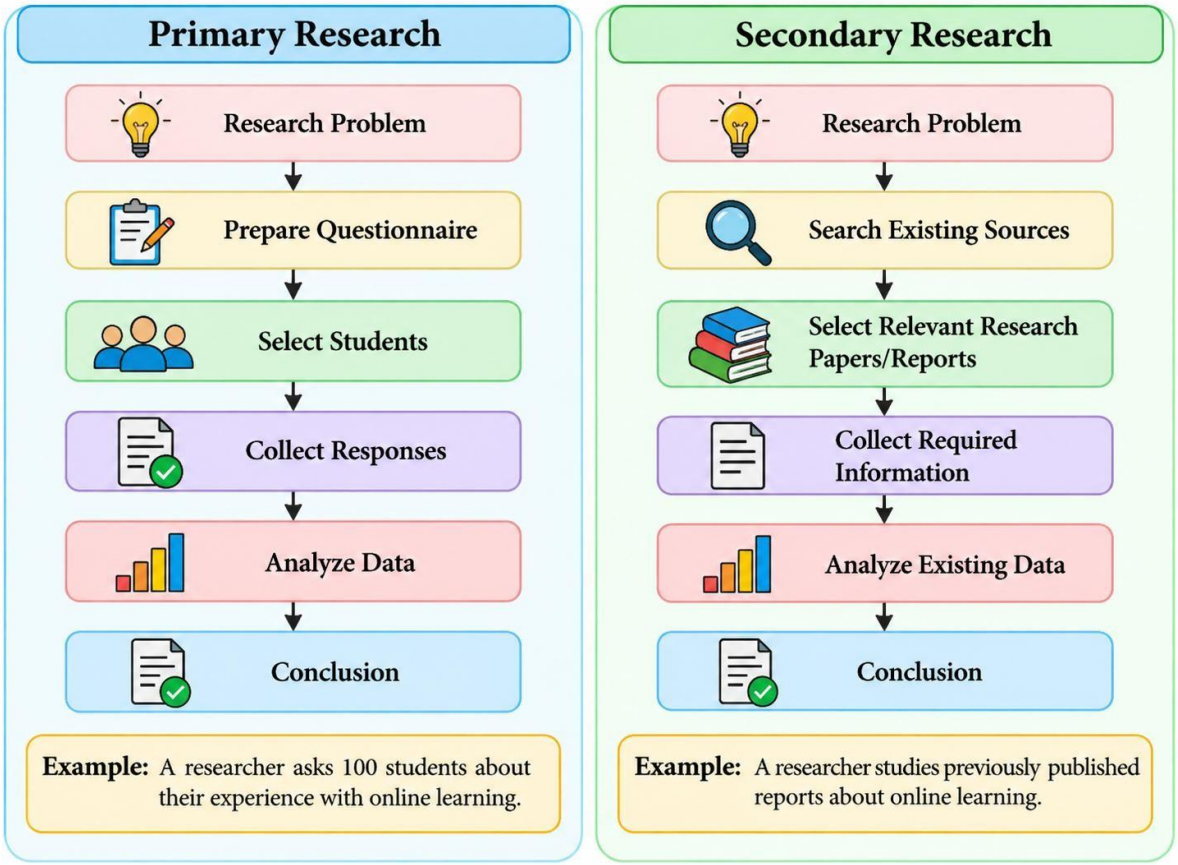
This information is **secondary data** because it was already collected and published by another source.

Difference Between Primary and Secondary Research

Sr. No.	Primary Research	Secondary Research
1	Collects original data	Uses existing data
2	Data is collected directly by the researcher	Data is collected by another person or organization
3	First-hand information	Second-hand information
4	Designed for a specific research problem	Originally collected for another purpose or broader use
5	Surveys, interviews and experiments are common methods	Books, journals, reports and websites are common sources
6	Generally requires more time	Generally requires less time
7	Can be expensive	Usually less expensive
8	Researcher has greater control over data collection	Researcher has limited control over how the original data was collected
9	Data can be highly specific to the study	Data may not exactly match the current research requirement
10	Example: Surveying 100 students	Example: Using a published education report

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Primary vs Secondary Research – Simple Illustration



Suitable Examples

Example 1 – Mobile Application

Research Question:
What features do users expect in a food delivery application?

Primary Research:
Conduct a survey of 200 users and ask them about their preferred features.

Secondary Research:
Study existing research papers, market reports and published articles about food delivery applications.

Example 2 – Artificial Intelligence

Research Question:
How are students using Artificial Intelligence tools for education?

Primary Research:
Interview students and collect their responses.

Secondary Research:
Study existing research papers and reports about AI in education.

Example 3 – E-Commerce

Research Question:
What factors influence online shopping?

Primary Research:
Conduct a questionnaire among online shoppers.

Secondary Research:
Use previously published market research reports and academic studies.

Advantages and Limitations

Primary Research

Advantages

- Provides original data.
- Data can be collected according to the research objectives.
- Researcher has control over the data collection process.
- Can provide current and specific information.

Limitations

- Can be time-consuming.
- May require more financial resources.
- Requires planning and participant cooperation.
- Data collection can be difficult for a large population.

Secondary Research

Advantages

- Saves time.
- Usually less expensive.
- Large amounts of information may be available.
- Useful for literature review and understanding previous research.

Limitations

- Data may be outdated.
- Information may not exactly match the research problem.
- Researcher has no control over the original data collection.
- Reliability depends on the quality of the original source.

Comparison Based on Research Scenario

Consider a research project:

"Study of Social Media Usage Among College Students."

Research Activity	Type	Example
Creating a questionnaire	Primary	Asking students about daily social media usage
Conducting interviews	Primary	Interviewing students
Observing user behaviour	Primary	Recording usage behaviour with appropriate consent
Reading research papers	Secondary	Studying previous research
Using government statistics	Secondary	Using published demographic data
Studying published reports	Secondary	Using social media usage reports

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Practical Activity

Task 1 – Primary Research

Create a questionnaire containing at least **10 questions** on:

"Use of Artificial Intelligence Tools Among College Students."

Collect responses from at least **20 students**.

Record the data in Excel.

Example columns:

Student ID	Course	AI Tool Used	Frequency	Purpose	Satisfaction
1	M.Sc. CA	ChatGPT	Daily	Study	High
2	M.Sc. CA	Gemini	Weekly	Coding	Medium
3	M.Sc. CA	ChatGPT	Daily	Assignments	High

The values above are only illustrative. Use actual collected responses in the practical.

Task 2 – Secondary Research

Find at least **5 reliable secondary sources** related to AI in education.

Record them in the following table:

Sr. No.	Source	Type of Source	Main Information
1	Research Paper	Journal	AI applications in education
2	Government Report	Report	AI/education statistics
3	Academic Book	Book	AI concepts
4	Research Article	Journal	Student use of AI

5	Institutional Report	Report	AI adoption

Data Analysis

After collecting primary data, use **MS Excel** to:

1. Calculate the number of students using AI tools.
2. Calculate usage frequency.
3. Categorize the purposes of AI usage.
4. Calculate percentages.
5. Create a **bar chart or pie chart**.
6. Interpret the results.

Example

If 20 students are surveyed:

AI Usage Frequency	Number of Students
Daily	10
Weekly	6
Occasionally	4

The researcher can create a bar chart to visualize the results.

Note: These numbers are only an example. Use actual survey results in the submitted practical.

Primary and Secondary Research in Computer Applications

Both methods are useful in computer application research.

Computer Application Research Activity	Research Type
User survey	Primary Research
Software usability testing	Primary Research
User interview	Primary Research
System experiment	Primary Research
Existing research paper analysis	Secondary Research
Studying software documentation	Secondary Research

Using published datasets	Secondary Research
Studying industry reports	Secondary Research
Reviewing previous projects	Secondary Research

Result

Primary and secondary research were studied and differentiated based on their data sources, collection methods, purpose, cost, time and level of researcher control. Primary research was identified as the collection of original first-hand data, while secondary research was identified as the use of existing information from previously published or collected sources. Suitable examples related to Computer Applications were also studied.

Conclusion

Primary and secondary research are two important approaches to conducting systematic research. Primary research provides original and specific information collected directly by the researcher, whereas secondary research uses existing information obtained from reliable sources. Primary research is useful when current and specific data is required, while secondary research is useful for understanding existing knowledge, previous studies and background information. In computer applications, both methods can be combined to conduct effective research, such as studying user requirements through surveys and reviewing existing research papers to understand previous work.

Practical No. 3

Title

Illustrate Basic, Applied and Translational Research and Differentiate Among Them with Suitable Example

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Aim

To study the concepts of Basic Research, Applied Research, and Translational Research, differentiate among them, and understand their applications in the field of Computer Applications through suitable examples.

Objectives

1. To understand the concept of Basic Research.
2. To understand the concept of Applied Research.
3. To study Translational Research and its significance.
4. To differentiate between Basic, Applied, and Translational Research.
5. To identify suitable examples of each research type in Computer Applications.

Requirements

Hardware

- Computer/Laptop
- Internet Connection

Software

- Microsoft Word / Google Docs
- Google Chrome / Microsoft Edge

Resources

- Google Scholar
- IEEE Xplore
- ACM Digital Library

- Research Methodology Textbook
- Research Journals

Practical Activity

Step 1: Study Basic Research

Study the meaning, objectives, and characteristics of Basic Research.

Understand:

- Purpose
- Characteristics
- Advantages
- Applications

Collect one real-life example related to Computer Applications.

Step 2: Study Applied Research

Study how Applied Research uses existing knowledge to solve practical problems.

Understand:

- Purpose
- Characteristics
- Applications
- Advantages

Collect one example from the software industry.

Step 3: Study Translational Research

Study how Translational Research converts scientific discoveries into practical technologies.

Understand:

- Purpose
- Process
- Applications
- Advantages

Collect one example related to Artificial Intelligence or Software Development.

Step 4: Compare the Three Types of Research

Parameter	Basic Research	Applied Research	Translational Research
Objective	Generate new knowledge	Solve practical problems	Convert research into real-world applications
Nature	Theoretical	Practical	Implementation
Outcome	New theories	New products or solutions	Commercial applications
Beneficiaries	Researchers	Industries	Society
Example	AI Algorithm Development	Spam Detection System	AI Chatbot for Healthcare

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Step 5: Select a Research Topic

Research Topic: Artificial Intelligence in Healthcare

Classify the topic into the following research categories:

Basic Research

Developing a new Deep Learning algorithm for detecting cancer cells from medical images with higher accuracy.

Applied Research

Developing an AI-based disease prediction system that analyzes patient data to detect diabetes and heart disease.

Translational Research

Deploying the AI disease prediction system in hospitals to assist doctors in diagnosis and patient care.

Step 6: Prepare a Short Report

Title

Artificial Intelligence in Healthcare: An Example of Basic, Applied and Translational Research

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Introduction

Artificial Intelligence (AI) has become one of the most important technologies in the healthcare sector. AI enables computers to perform tasks that normally require human intelligence, such as disease diagnosis, medical image analysis, drug discovery, patient monitoring, and personalized treatment. Technologies such as Machine Learning (ML), Deep Learning (DL), and Natural Language Processing (NLP) help healthcare professionals improve the quality, speed, and accuracy of medical services.

Research in Artificial Intelligence can be classified into three categories: **Basic Research**, **Applied Research**, and **Translational Research**. These research types work together to transform scientific discoveries into practical healthcare solutions that improve patient care and save lives.

Basic Research

Basic Research, also known as Fundamental Research, aims to generate new scientific knowledge without focusing on immediate practical applications. It seeks to discover new theories, algorithms, and techniques that can be used in future innovations.

Example

Researchers develop a new Deep Learning algorithm capable of detecting cancer cells in microscopic images with greater accuracy than existing models. The algorithm is trained and tested using medical image datasets to improve its performance. At this stage, the research focuses on expanding scientific knowledge rather than direct clinical use.

Outcome

- Development of advanced AI algorithms.
- Improved understanding of medical image processing.
- Foundation for future healthcare technologies.

Applied Research

Applied Research uses the knowledge generated through Basic Research to solve practical problems. It focuses on developing systems and technologies that can be used in real-world situations.

Example

An AI-based disease prediction system is developed to analyze patient information such as age, blood pressure, blood sugar, cholesterol levels, and medical history. The system predicts diseases such as diabetes and heart disease, helping doctors make faster and more accurate decisions.

Outcome

- AI-based diagnostic software.
- Faster disease prediction.
- Improved healthcare services.
- Better clinical decision-making.

Translational Research

Translational Research bridges the gap between scientific discoveries and practical implementation. It converts successful research outcomes into products and technologies that directly benefit society.

Example

The AI disease prediction system is implemented in hospitals and healthcare centers. Doctors use the system to analyze patient reports, assist in diagnosis, monitor patient health remotely, and provide personalized treatment recommendations. Patients can also interact with AI-powered healthcare chatbots to receive preliminary guidance and schedule appointments.

Outcome

- AI systems deployed in hospitals.
- Remote patient monitoring.
- Improved healthcare accessibility.
- Better patient outcomes.

Comparison of Basic, Applied and Translational Research

Feature	Basic Research	Applied Research	Translational Research
Purpose	Discover new knowledge	Solve practical problems	Implement research findings
Focus	Theory	Practical solution	Real-world implementation
Nature	Scientific	Application-oriented	Product-oriented
Time Required	Long-term	Medium-term	Short to Medium
Output	Algorithms and theories	Software systems	Commercial products
Example	New AI Algorithm	Disease Prediction System	AI Healthcare Platform

Applications in Computer Applications

Basic Research

- Artificial Intelligence Algorithms
- Machine Learning Models
- Quantum Computing
- Data Structures and Algorithms

Applied Research

- Cybersecurity Applications
- Mobile Applications
- Software Engineering
- Data Analytics
- Cloud Computing

Translational Research

- AI Healthcare Systems
- Smart Cities
- Intelligent Chatbots
- Autonomous Vehicles
- Industrial Automation

Conclusion

Basic Research, Applied Research, and Translational Research are interconnected stages of innovation. Basic Research develops new scientific knowledge, Applied Research transforms this knowledge into practical solutions, and Translational Research implements these solutions in real-world environments.

In Computer Applications, these research methodologies play a significant role in developing Artificial Intelligence systems, cybersecurity solutions, cloud computing platforms, and intelligent software applications. Understanding these research types enables students to conduct high-quality research and develop innovative technologies that solve real-world problems and contribute to society.

Expected Output

After completing this practical, the student should be able to:

- Explain Basic, Applied, and Translational Research.
- Differentiate among the three research types.
- Identify suitable examples in Computer Applications.
- Classify research problems based on research type.
- Prepare a structured research report.

Result

The concepts of Basic Research, Applied Research, and Translational Research were successfully studied and compared using the example of **Artificial Intelligence in Healthcare**. The practical demonstrated how scientific knowledge is generated, applied to solve practical healthcare problems, and finally implemented as AI-powered healthcare systems that improve diagnosis, treatment, and patient care.

Practical No. 4

Title

Present Qualitative, Quantitative and Mixed Research and Differentiate Among Them

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Aim

To understand **Qualitative, Quantitative and Mixed Research**, study their characteristics and methods, and differentiate among them with suitable examples.

Objectives

1. To understand the meaning of qualitative research.
2. To understand the meaning of quantitative research.
3. To understand the concept of mixed research.
4. To identify the methods used in each type of research.
5. To differentiate qualitative, quantitative and mixed research.
6. To understand their applications in Computer Applications.

Requirements

Hardware

- Computer/Laptop
- Internet connection

Software/Tools

- MS Word / Google Docs
- MS Excel / Google Sheets
- Web browser
- MS PowerPoint

Reference Sources

- Research methodology textbooks
- Academic research papers
- Journals
- Reliable websites
- Published research reports

Introduction

Research can be conducted using different approaches depending on the type of information required. The three important approaches are **Qualitative Research, Quantitative Research and Mixed Research**.

Qualitative Research focuses on understanding opinions, experiences, perceptions and behaviours using non-numerical data.

Quantitative Research focuses on collecting and analyzing numerical data using measurements, statistics and mathematical techniques.

Mixed Research combines both qualitative and quantitative approaches in a single research study to obtain a broader understanding of the research problem.

These approaches are widely used in fields such as computer applications, information technology, education, healthcare, business and social sciences.

Qualitative Research

Definition

Qualitative Research is a research approach used to understand people's **experiences, opinions, perceptions, attitudes and behaviours** using non-numerical information.

Common Methods

- Interviews

- Focus groups
- Observation
- Case studies
- Open-ended questionnaires
- Document analysis

Example

Research Topic:
"Students' Experiences with Online Learning."

A researcher interviews 20 students and asks them about:

- Their learning experience
- Problems faced during online classes
- Their preferences
- Suggestions for improvement

The researcher analyzes the responses to identify common themes and opinions.

Quantitative Research

Definition

Quantitative Research is a research approach that collects and analyzes **numerical data** to measure variables, identify relationships and test hypotheses.

Common Methods

- Surveys with closed-ended questions
- Experiments
- Statistical analysis
- Structured observations
- Measurements

Example

Research Topic:
"Online Learning Usage Among Students."

A researcher surveys 200 students and collects information such as:

- Number of hours spent online

- Number of classes attended
- Test scores
- Satisfaction rating from 1 to 5

The researcher analyzes the numerical data using percentages, averages, graphs and statistical tests.

Mixed Research

Definition

Mixed Research combines **qualitative and quantitative research methods** in the same study.

It provides both numerical measurements and detailed information about people's experiences or opinions.

Common Methods

- Surveys + Interviews
- Questionnaires + Focus Groups
- Statistical analysis + Thematic analysis
- Experiments + Interviews

Example

Research Topic:
"Effectiveness of Online Learning Among College Students."

A researcher:

1. Surveys 200 students to measure satisfaction and performance.
2. Interviews 20 students to understand their experiences and problems.
3. Combines the numerical and qualitative findings to reach a conclusion.

Simple Illustration



Difference Between Qualitative, Quantitative and Mixed Research

Sr. No.	Qualitative Research	Quantitative Research	Mixed Research
1	Uses non-numerical data	Uses numerical data	Uses both types of data
2	Focuses on opinions and experiences	Focuses on measurement	Combines measurement and experiences
3	Interviews and observations are common	Surveys and experiments are common	Uses both methods
4	Data is usually descriptive	Data is statistical	Data is descriptive and statistical
5	Smaller samples are often used	Larger samples are often used	Sample size depends on the study
6	Thematic analysis is commonly used	Statistical analysis is commonly used	Uses both types of analysis
7	Answers questions such as "Why?" and "How?"	Answers questions such as "How much?" and "How many?"	Can answer both types of questions
8	Flexible research design	More structured research design	Combines approaches
9	Example: Student interviews	Example: Student survey	Example: Survey + interviews

Advantages and Limitations

Qualitative Research

Advantages

- Provides detailed information.
- Helps understand opinions and experiences.
- Provides deeper understanding of human behaviour.
- Flexible data collection.

Limitations

- Analysis can be time-consuming.
- Results can be difficult to generalize.
- Usually involves smaller samples.
- Interpretation may require careful researcher judgment.

Quantitative Research

Advantages

- Produces measurable results.
- Suitable for statistical analysis.
- Can use large samples.
- Results can be presented using graphs and tables.

Limitations

- May not provide detailed explanations of opinions.
- Requires properly designed measurements.
- Some human experiences may be difficult to represent numerically.

Mixed Research

Advantages

- Combines strengths of both approaches.
- Provides numerical and detailed information.

- Gives a broader understanding of the research problem.
- Useful for complex research problems.

Limitations

- Requires more time and resources.
- Data analysis can be more complex.
- Requires knowledge of both qualitative and quantitative methods.

Examples in Computer Applications

Research Topic	Qualitative	Quantitative	Mixed
Software usability	User interviews	Usability scores	Interviews + scores
Mobile applications	User opinions	Number of downloads/usage	Opinions + usage data
AI in education	Student experiences	AI usage frequency	Experiences + usage statistics
Cybersecurity awareness	Interviews	Awareness scores	Interviews + survey
Online learning	Student experiences	Test scores	Experiences + test scores

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Practical Activity

Task 1 – Qualitative Research

Conduct interviews with **at least 10 students** about:

"Experience of Students Using Artificial Intelligence Tools."

Record their responses and identify common themes.

Example themes:

- Academic assistance
- Coding support
- Assignment preparation
- Information searching
- Time saving

Task 2 – Quantitative Research

Prepare a questionnaire and collect responses from **at least 20 students**.

Example variables:

Student ID	AI Usage Frequency	Hours/Week	Satisfaction (1–5)
1	Daily	8	5
2	Weekly	4	4
3	Occasionally	2	3

The values above are illustrative. Use actual collected data in the practical.

Create:

- Bar chart
- Pie chart
- Percentage calculation
- Average satisfaction score

Task 3 – Mixed Research

Combine the two activities.

Quantitative Data

Measure:

- Frequency of AI usage
- Hours of usage
- Satisfaction score

Qualitative Data

Study:

- Student opinions
- Benefits
- Problems
- Suggestions

Finally, compare both findings and prepare a combined conclusion.

Expected Output

Activity

A small study was conducted on **"Use of Artificial Intelligence Tools Among College Students."**

1. Qualitative Activity

Interviews were conducted with students to understand their experiences, opinions and difficulties while using AI tools.

Main findings:

- AI helps in understanding difficult topics.
- Students use AI for coding and assignments.
- Some students were concerned about incorrect information.
- Students suggested using AI as a learning support tool.

2. Quantitative Activity

A questionnaire was used to collect numerical data from **20 students**.

AI Usage	Students
Daily	10
Weekly	6
Occasionally	4
Total	**20**

The collected data was organized and analyzed using Excel.

3. Mixed Research Activity

The qualitative interview findings were combined with the quantitative survey results. The numerical data showed **how frequently students use AI**, while the interviews explained **why and how they use it**.

Comparison

Research Type	Main Data	Method
Qualitative	Opinions and experiences	Interviews
Quantitative	Numerical data	Questionnaire
Mixed	Both types	Survey + Interview

4. Result

Qualitative, quantitative and mixed research methods were successfully studied and differentiated through a practical study of AI usage among students.

Result

Qualitative, Quantitative and Mixed Research were studied and compared. Qualitative research was identified as an approach based mainly on non-numerical information such as opinions and experiences, quantitative research as an approach based on numerical measurement and statistical analysis, and mixed research as a combination of both approaches. Their methods, advantages, limitations and applications in Computer Applications were also studied.

Conclusion

Qualitative, quantitative and mixed research provide different approaches for investigating research problems. **Qualitative research** helps understand experiences, opinions and behaviour, while **quantitative research** provides measurable and statistically analyzable results. **Mixed research** combines both approaches to obtain a more comprehensive understanding of a research problem. In Computer Applications, these methods can be applied to areas such as software usability, artificial intelligence, cybersecurity, mobile applications and online learning.

Practical No. 5

Title

Illustrate Various Steps of Literary Research: Narrative Review, Systematic Review, and Meta-analysis, and Identify the Differences Among Them

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Aim

To study the various steps involved in literary research, understand Narrative Review, Systematic Review, and Meta-analysis, and identify the differences among them using a research topic related to Computer Applications.

Objectives

1. To understand the concept of literature review.
2. To study the steps involved in conducting a literature review.
3. To understand Narrative Review, Systematic Review, and Meta-analysis.
4. To compare the three review methods.
5. To perform a literature review using a Computer Application research topic.

Requirements

Hardware

- Computer/Laptop
- Internet Connection

Software

- Google Chrome / Microsoft Edge
- Microsoft Word / Google Docs

Resources

- Google Scholar
- IEEE Xplore

- ACM Digital Library
- ResearchGate
- Scopus
- Research Methodology Textbook

Practical Activity

Step 1: Select a Research Topic

Choose the research topic:

Artificial Intelligence in Education

Step 2: Search Research Papers

Search at least five research papers using Google Scholar or IEEE Xplore.

Selected Papers

1. Artificial Intelligence in Education: Challenges and Opportunities
2. ChatGPT in Higher Education
3. Machine Learning for Student Performance Prediction
4. Intelligent Tutoring Systems Using AI
5. AI-Based Personalized Learning Platforms

Step 3: Prepare a Narrative Review

Read all five research papers and prepare a general summary of the findings without following any fixed protocol.

Step 4: Prepare a Systematic Review

Perform a structured review by following these steps:

- Define the research question.
- Select databases.
- Define inclusion and exclusion criteria.

- Select relevant research papers.
- Analyze the selected studies.
- Summarize the findings.

Step 5: Prepare a Meta-analysis

Collect quantitative data from the selected studies and compare their results statistically.

Example:

- Accuracy of AI prediction models
- Student performance improvement (%)
- Learning efficiency
- Model accuracy

Summarize the combined findings.

Step 6: Compare Narrative Review, Systematic Review and Meta-analysis

Parameter	Narrative Review	Systematic Review	Meta-analysis
Purpose	General summary	Structured review	Statistical analysis
Search Method	Flexible	Predefined	Uses systematic review data
Selection Criteria	No fixed criteria	Strict criteria	Strict criteria
Bias	Higher	Lower	Lowest
Data	Qualitative	Qualitative + Quantitative	Quantitative
Result	Overview	Evidence-based conclusion	Combined statistical result

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Step 7: Prepare a Short Report

Title

Literature Review on Artificial Intelligence in Education Using Narrative Review, Systematic Review, and Meta-analysis

Introduction

A literature review is a systematic process of collecting, evaluating, and summarizing existing research on a specific topic. It helps researchers understand previous studies, identify research gaps, and avoid duplication of work. In Computer Applications, literature reviews are essential before starting software development, Artificial Intelligence (AI), Machine Learning (ML), Data Science, or Cybersecurity projects.

There are three major approaches to literature review:

- Narrative Review
- Systematic Review
- Meta-analysis

Each approach has its own methodology, advantages, and applications.

Narrative Review

A Narrative Review provides a descriptive summary of previous studies on a research topic. The researcher selects relevant papers and summarizes their findings without following a strict selection protocol.

Example

Five research papers related to **Artificial Intelligence in Education** were reviewed. The studies discussed the use of AI in intelligent tutoring systems, personalized learning, student performance prediction, and automated assessment. Most researchers concluded that AI improves learning efficiency, enhances student engagement, and supports teachers in academic activities.

Advantages

- Easy to conduct
- Provides broad understanding
- Suitable for exploratory research

Limitations

- Subjective
- Higher risk of bias

- No statistical analysis

Systematic Review

A Systematic Review follows a structured and transparent methodology to identify, evaluate, and summarize research studies.

Research Question

How does Artificial Intelligence improve teaching and learning in higher education?

Search Databases

- Google Scholar
- IEEE Xplore
- ACM Digital Library

Inclusion Criteria

- Published after 2020
- Peer-reviewed journals
- AI in Education
- English language

Exclusion Criteria

- Duplicate papers
- Conference abstracts only
- Non-English publications

Findings

After reviewing the selected papers, it was observed that AI-based learning systems improve student performance, automate assessments, personalize learning experiences, and reduce teachers' workload.

Advantages

- Transparent methodology
- Reliable findings
- Evidence-based conclusions

Limitations

- Time-consuming
- Requires strict planning

Meta-analysis

Meta-analysis is a statistical technique used to combine numerical results from multiple research studies to obtain a more accurate conclusion.

Example

Five AI education studies reported student performance improvement after implementing AI-based learning systems.

Study	Improvement (%)
Study 1	18%
Study 2	22%
Study 3	20%
Study 4	24%
Study 5	21%

Combined Result

Average improvement:

(18 + 22 + 20 + 24 + 21) ÷ 5 = 21%

This indicates that AI-based educational systems improved student performance by approximately **21%** across the selected studies.

Advantages

- Highly reliable
- Statistical evidence
- Reduces uncertainty

Limitations

- Requires quantitative data
- Complex statistical analysis

Comparison of the Three Review Methods

Feature	Narrative Review	Systematic Review	Meta-analysis
Purpose	General understanding	Evidence-based review	Statistical conclusion
Method	Flexible	Structured	Statistical
Bias	High	Low	Very Low
Data Type	Qualitative	Qualitative & Quantitative	Quantitative
Reliability	Moderate	High	Very High
Time Required	Low	Medium	High

Applications in Computer Applications

Narrative Review

- AI
- Machine Learning
- Cybersecurity
- Web Development

Systematic Review

- Software Engineering
- Cloud Computing
- Blockchain
- Data Science

Meta-analysis

- AI Model Performance
- Medical Image Classification
- Network Security
- Software Testing Accuracy

Conclusion

A literature review is one of the most important stages of research because it helps researchers understand existing knowledge and identify research gaps. Among the three review methods, the **Narrative Review** provides a broad overview of previous studies, the **Systematic Review** offers an organized and evidence-based evaluation, and the **Meta-analysis** combines quantitative findings to produce statistically reliable conclusions.

For Computer Application students, these review techniques are valuable when conducting research in Artificial Intelligence, Machine Learning, Software Engineering, Cybersecurity, and Data Science. Selecting the appropriate review method improves the quality, reliability, and credibility of research projects.

Expected Output

After completing this practical, the student should be able to:

- Explain the concept of literature review.
- Conduct a Narrative Review.
- Perform a Systematic Review.
- Understand the basics of Meta-analysis.
- Differentiate among the three review methods.
- Prepare a structured literature review report using research databases.

Result

The literature review process was successfully performed using the topic **"Artificial Intelligence in Education."** Narrative Review, Systematic Review, and Meta-analysis were studied and compared based on their methodology, data collection process, reliability, and outcomes. The practical demonstrated that while all three methods contribute to research, **Systematic Review and Meta-analysis provide stronger evidence and more reliable conclusions**, making them highly valuable for research in **Computer Applications**.

Practical No. 6

Title: Illustration of Various Terms Related to Research Designs with the Help of Case Studies in Computer Applications

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Aim: To understand and illustrate various terms related to research design by studying different case studies in the field of Computer Applications.

Objective:

1. To understand the meaning and importance of research design.
2. To identify different types of research designs.
3. To understand how research designs are applied in Computer Applications.
4. To differentiate between qualitative, quantitative, experimental, descriptive, and case-study research.
5. To develop the ability to select a suitable research design for a given problem.

Requirement:

1. Computer/Laptop
2. Internet connection
3. MS Word/Google Docs
4. Research papers or case studies related to Computer Applications

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Theory / Background:

A **research design** is a systematic plan that describes how a research study will be conducted. It specifies the methods of collecting, analyzing, and interpreting data.

Important terms related to research design include:

- **Research Problem:** The issue or question that the researcher wants to investigate.
- **Research Question:** A specific question that the research attempts to answer.
- **Hypothesis:** A testable statement or prediction about the relationship between variables.
- **Variable:** A characteristic that can change or be measured.
- **Independent Variable:** The factor that is changed or controlled by the researcher.
- **Dependent Variable:** The factor that is measured as an outcome.
- **Population:** The complete group about which the researcher wants to draw conclusions.
- **Sample:** A smaller group selected from the population.
- **Data Collection:** The process of gathering information for the research.

- **Research Method:** The technique used to conduct the research.
- **Research Design:** The overall plan or structure of the research.

Types of Research Designs:

Research Design	Meaning	Computer Application Example
Descriptive	Describes characteristics or existing situations	Study of students' use of learning applications
Experimental	Tests the effect of one variable on another	Testing whether a new algorithm improves search speed
Qualitative	Studies opinions, experiences, and behaviors	Interviews about students' experiences with AI tools
Quantitative	Uses numerical data and statistical analysis	Measuring website performance using response time
Case Study	Detailed investigation of a particular case	Study of a university's implementation of an LMS
Survey	Collects information from a group using questionnaires	Survey of students' cybersecurity awareness
Survey	Compares two or more systems, methods, or groups	Comparing two programming languages for development time

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Case Study 1: E-Learning Application

Research Problem:

Students are increasingly using online learning applications, but their effectiveness may differ among students.

Research Question:

How does the use of an e-learning application affect students' learning performance?

Research Design:

Quantitative and descriptive research design.

Variables:

- Independent Variable: Usage of the e-learning application
- Dependent Variable: Student test scores

Sample:

50 students using an e-learning application.

Data Collection:

Collect application usage time and test scores through questionnaires and academic assessments.

Expected Analysis:

Compare the amount of application usage with students' test performance using tables, graphs, or statistical methods.

Case Study 2: Website Performance

Research Problem:

A website has a slow response time during periods of high traffic.

Research Question:

Does optimizing the database improve website response time?

Research Design:

Experimental research design.

Variables:

- Independent Variable: Database optimization
- Dependent Variable: Website response time

Activity:

1. Measure the website response time before optimization.
2. Optimize the database.
3. Measure response time after optimization.
4. Compare the results.

Expected Analysis:

Determine whether database optimization changes the website's response time.

Case Study 3: Student Cybersecurity Awareness

Research Problem:

Students may not have sufficient knowledge about common cybersecurity threats.

Research Question:

What is the level of cybersecurity awareness among college students?

Research Design:

Survey research design.

Activity:

1. Prepare a questionnaire containing questions about passwords, phishing, malware, and safe browsing.
2. Select a sample of students.
3. Distribute the questionnaire.
4. Collect responses.
5. Calculate percentages and represent the findings using charts.

Expected Analysis:

Identify the general level of cybersecurity awareness among the selected students.

Practical Activity:

Activity: Identify the Appropriate Research Design

Consider the following computer application problems:

1. **Comparing the performance of Java and Python programs.**
2. **Studying students' opinions about online education.**
3. **Testing whether a new sorting algorithm reduces execution time.**
4. **Studying how a company implemented a new ERP system.**
5. **Finding the percentage of students who use cloud storage.**

For each problem, identify:

- Research problem
- Research question
- Research design
- Variables (where applicable)
- Population
- Sample
- Data collection method
- Expected result

Expected Output:

A completed table identifying the appropriate research design for each case study, along with the research problem, variables, sample, data collection method, and expected outcome.

Result:

The various terms related to research designs were studied and illustrated using case studies from Computer Applications. Different research designs such as **descriptive, experimental, qualitative, quantitative, survey, comparative, and case-study designs** were identified and applied to suitable research problems.

Conclusion:

Research design provides a structured approach for conducting research. Selecting an appropriate research design helps researchers collect relevant data, analyze the problem systematically, and obtain meaningful results in the field of Computer Applications.

Practical No. 7

Title

Describe Cross-Sectional Study with Specific Case Study

Aim

To understand the concept of literature review under peer learning, study existing research related to peer learning, identify major findings and research gaps, and understand how literature review supports the development of a research problem.

Objectives

- To understand the meaning and importance of a literature review.
- To understand the concept of peer learning.
- To study existing research papers related to peer learning.
- To identify similarities and differences among previous studies.
- To identify important findings from existing literature.
- To identify research gaps in peer learning.
- To understand how literature review helps in formulating research questions.
- To use digital tools for collecting, organizing and analyzing research literature.

Requirements

Hardware

- Computer/Laptop
- Internet connection
- Printer (optional)

Software

- Web browser
- MS Word / Google Docs
- MS Excel / Google Sheets
- MS PowerPoint
- Google Scholar or other academic databases
- Reference management tools such as Zotero/Mendeley (optional)

Resources

- Research papers
- Review papers
- Systematic reviews
- Meta-analysis pape

Practical Activity

Step 1: Introduction

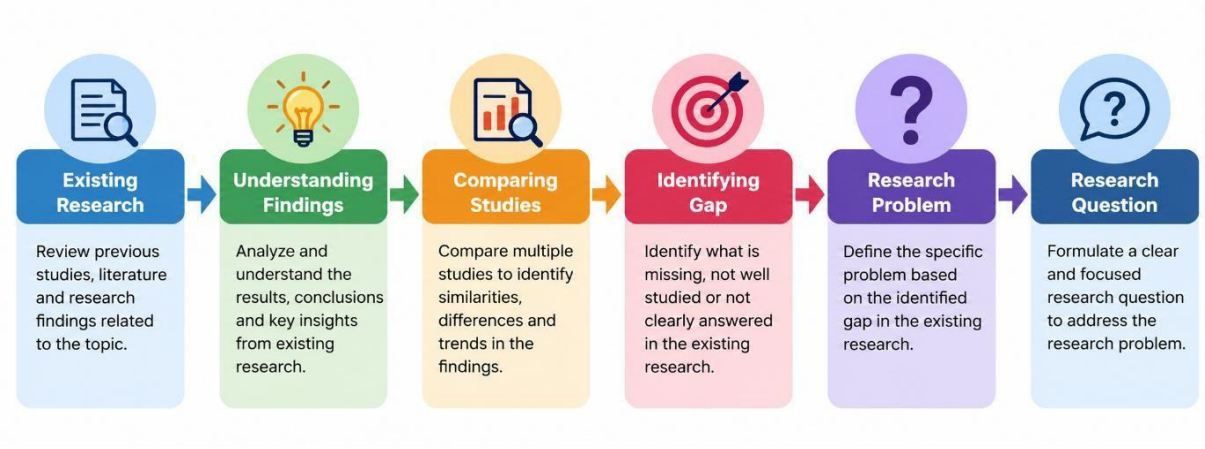
A literature review is a systematic study and analysis of previously published research related to a particular topic. It helps a researcher understand what has already been studied, what methods have been used, what results have been obtained and what research gaps still exist.

Peer learning is a learning approach in which students learn with and from other students. It can include peer tutoring, peer teaching, peer assessment, collaborative learning and peer instruction.

Research on peer learning has developed through different forms and educational contexts. Topping's review of peer learning discussed forms such as peer tutoring, cooperative learning and peer assessment, and also described developments in the use of information technology in peer learning.

Research reviews have also examined the effectiveness of peer learning. For example, a 2017 meta-analysis examined 29 Peer Instruction studies involving 4,738 participants and investigated learning gains and possible moderating factors.

Step 2: Create a Research Timeline



Step 3: Meaning of Literature Review

A literature review is an organized examination of previously published research related to a particular topic.

- Searching for relevant literature
- Selecting appropriate sources
- Reading research papers

- Understanding research methodology
- Extracting important findings
- Comparing different studies
- Identifying research gaps
- Organizing information
- Referencing the sources
- Developing a research direction

Step 4: Understanding Peer Learning

Peer learning is an educational approach where learners participate in learning activities with other learners who have a similar or related level of knowledge or experience**.**

Common forms of peer learning include:

1. Peer Tutoring
2. Peer Teaching
3. Peer Assessment
4. Peer Instruction
5. Collaborative Learning
6. Online Peer Learning

Step 5: Importance of Literature Review in Peer Learning

Literature review is important when studying peer learning because it helps researchers understand how peer learning has previously been implemented and evaluated.

Major purposes include:

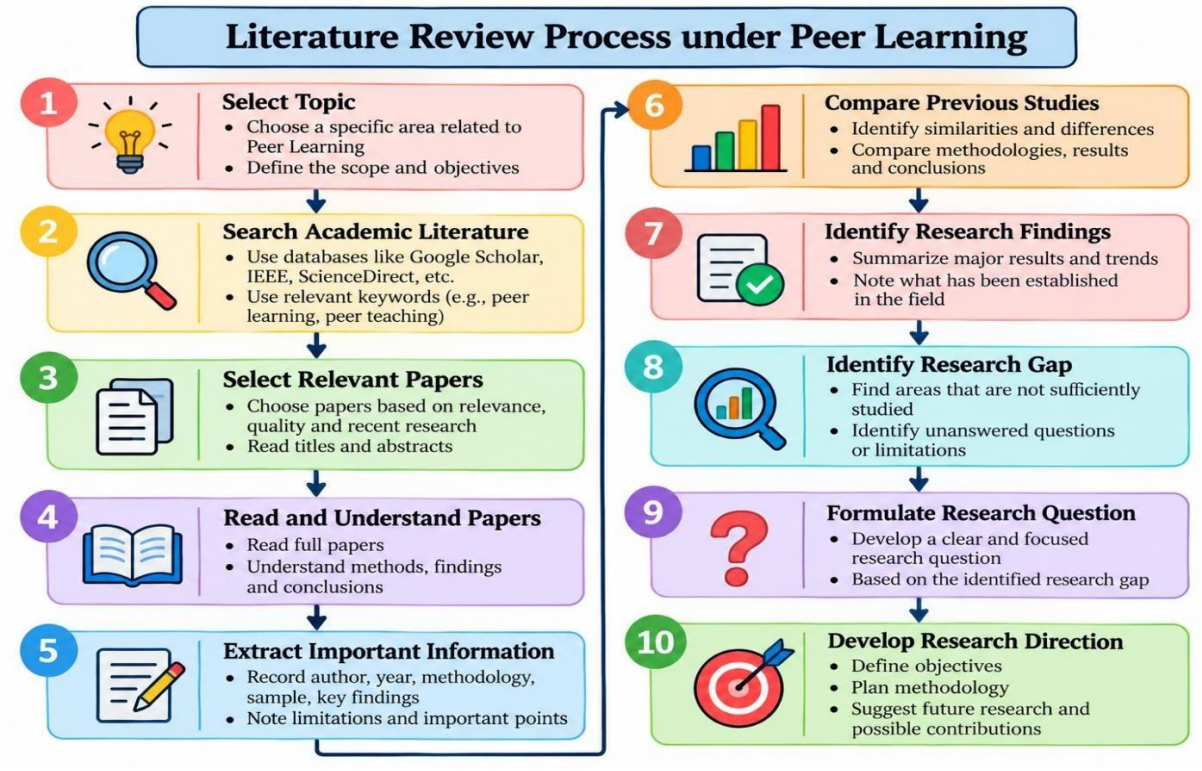
1. Understanding existing peer-learning models
2. Identifying commonly used teaching methods
3. Studying learning outcomes
4. Comparing traditional and peer-learning approaches
5. Identifying benefits and limitations
6. Understanding student experiences
7. Identifying research gaps
8. Developing new research questions
9. Selecting suitable research methodology

Historical Development of Peer Learning Research

Period	Development in Peer Learning
Early educational practices	Students learned through group activities and knowledge sharing
20th century	Peer tutoring and cooperative learning became more structured
1980s–1990s	Peer tutoring, cooperative learning and peer assessment received increased research attention
1990s–2000s	Peer learning expanded into higher education
2000s	Peer instruction and technology-supported learning developed
2010s	Systematic reviews and meta-analyses increasingly examined peer learning
2020s	Online, blended and technology-supported peer learning became important research areas
Present	Digital collaboration, AI-supported learning and computational analysis provide new research opportunities

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Literature Review Process under Peer Learning



Role of Technology in Literature Review

Modern computer applications can make literature review more efficient.

Useful technologies include:

- Google Scholar for searching research papers
- Excel for organizing papers
- Zotero/Mendeley for references
- Word/Google Docs for writing
- PowerPoint for presenting findings
- Python for bibliometric analysis
- Data visualization tools for analyzing publication trends
- AI tools for assisting with literature organization and summarization, while researchers must verify information against original sources

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Conclusion

Literature review is an essential part of the research process because it helps researchers understand existing knowledge, compare previous studies and identify research gaps. Peer learning provides a collaborative environment in which students can share knowledge, explain concepts and learn from one another. Existing research has examined different forms of peer learning, including peer tutoring, peer instruction, peer assessment and peer-assisted learning. Reviews and meta-analyses show that peer learning can have useful educational effects, although the strength and consistency of outcomes can vary by context and the outcome being measured.

Expected Output

After completing the practical, the student should be able to:

- Explain the concept and importance of literature review in research.
- Identify and summarize relevant research studies related to peer learning.
- Compare the findings, methodologies, and approaches used in different research studies.
- Identify research gaps from the existing literature.
- Understand how literature review helps in formulating research questions and objectives.
- Apply peer learning techniques such as discussion, peer teaching, and collaborative analysis during literature review.
- Organize and analyze research literature using digital tools such as Excel, Google Scholar, Word, and PowerPoint.
- Prepare a structured literature review report using digital tools.

Result

The literature review process under peer learning was studied successfully. Relevant research papers related to peer learning were identified, organized and compared according to research methodology, peer-learning type, findings and research gaps. The collected literature was digitally organized and analyzed using computer-based tools.

Practical No. 8

Title

Describe Randomized Controlled Trial (RCT)

Aim

To study the concept, methodology, and importance of Randomized Controlled Trials (RCTs) and understand their applications in research with a suitable example related to Computer Applications.

Objectives

1. To understand the concept of a Randomized Controlled Trial (RCT).
2. To study the steps involved in conducting an RCT.
3. To understand the advantages and limitations of RCTs.
4. To identify the applications of RCTs in healthcare and Computer Applications.
5. To perform a simple RCT case study using a Computer Application example.

Requirements

Hardware

- Computer/Laptop
- Internet Connection

Software

- Microsoft Word / Google Docs
- Google Chrome / Microsoft Edge
- Microsoft Excel (for recording observations)

Resources

- Research Methodology Textbook
- Google Scholar
- IEEE Xplore
- Research Journals
- Clinical Trial Articles

Practical Activity

Step 1: Study Randomized Controlled Trial (RCT)

Study the definition, purpose, characteristics, and importance of Randomized Controlled Trials.

Understand:

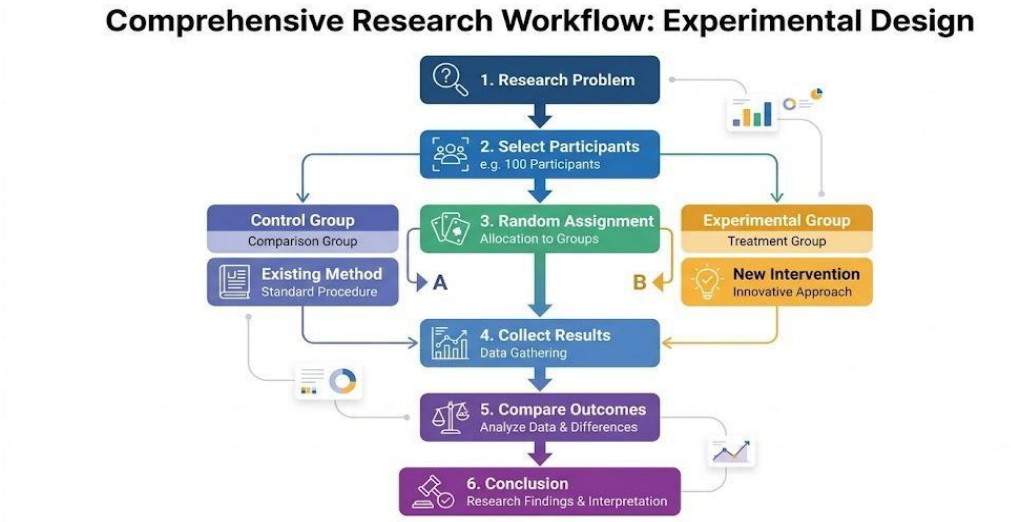
- Randomization
- Control Group
- Experimental Group
- Outcome Measurement
- Data Analysis

Step 2: Understand the Components of an RCT

Study the following components:

- Research Problem
- Research Participants
- Random Assignment
- Experimental Group
- Control Group
- Intervention
- Data Collection
- Statistical Analysis
- Conclusion

Step 3: Prepare the Flow of an RCT



Step 4: Conduct a Case Study

Research Topic

Effectiveness of an AI-Based Learning Platform on Student Performance

Research Problem

Does an AI-based learning platform improve students' academic performance compared to traditional learning?

Participants

60 undergraduate Computer Application students.

Randomization

Students are randomly divided into two groups.

Group	Number of Students	Learning Method
Control Group	30	Traditional Classroom Learning
Experimental Group	30	AI-Based Learning Platform

Intervention

The experimental group studies using an AI-powered personalized learning platform for four weeks.

The control group continues traditional classroom learning during the same period.

Data Collection

After four weeks, both groups appear for the same examination.

Group	Average Marks
Control Group	72
Experimental Group	84

Observation

The experimental group achieved higher average marks than the control group.

Step 5: Analyze the Results

Calculate the improvement.

Average Improvement:

84 − 72 = **12 Marks**

Percentage Improvement:

\[\frac{12}{72} \times 100 = 16.67\% \]

The AI-based learning platform improved students' performance by approximately **16.67%**.

Step 6: Prepare a Short Report

Title

Randomized Controlled Trial on the Effectiveness of an AI-Based Learning Platform

Introduction

A **Randomized Controlled Trial (RCT)** is one of the most reliable research methods used to evaluate the effectiveness of a treatment, intervention, or technology. Participants are randomly assigned to either an **experimental group**, which receives the intervention, or a

control group, which follows the existing method. Randomization minimizes bias and ensures that the results are scientifically valid.

Although RCTs are widely used in medicine and healthcare, they are also increasingly applied in **Computer Applications**, particularly in evaluating educational software, Artificial Intelligence systems, cybersecurity tools, and human-computer interaction.

Methodology

The study involved **60 undergraduate Computer Application students**.

Students were randomly assigned into two groups:

- **Control Group:** 30 students studied using traditional classroom methods.
- **Experimental Group:** 30 students used an AI-based personalized learning platform.

Both groups studied the same syllabus for four weeks and then completed the same examination.

Observations

Group	Students	Average Marks
Control Group	30	72
Experimental Group	30	84

The experimental group performed significantly better than the control group.

Data Analysis

The difference between the two groups was:

84 − 72 = **12 Marks**

Percentage improvement:

16.67%

This indicates that students who used the AI-based learning platform achieved better academic performance.

Advantages of Randomized Controlled Trial

- Reduces research bias.
- Produces reliable and accurate results.
- Allows fair comparison between two groups.
- Provides evidence-based conclusions.
- Considered the gold standard of experimental research.

Limitations of Randomized Controlled Trial

- Time-consuming.
- Expensive to conduct.
- Requires careful planning.
- Ethical issues may arise in some studies.
- Large sample sizes may be required.

Applications in Computer Applications

Randomized Controlled Trials can be used to evaluate:

- AI-Based Learning Platforms
- Educational Software
- Machine Learning Models
- Cybersecurity Systems
- Mobile Applications
- User Interface (UI/UX) Designs
- Recommendation Systems
- Intelligent Chatbots

Conclusion

Randomized Controlled Trials are one of the most effective research methods for evaluating the performance of new technologies and interventions. In this practical, an AI-based learning platform was compared with traditional classroom learning using an RCT approach. The results demonstrated that the AI-based platform significantly improved students' academic performance.

For Computer Application students, understanding Randomized Controlled Trials is valuable because they provide a scientific method for testing software systems, AI models, educational technologies, and digital applications before their implementation in real-world environments.

Expected Output

After completing this practical, the student should be able to:

- Explain the concept of Randomized Controlled Trials.
- Describe the steps involved in conducting an RCT.
- Understand the importance of randomization.
- Analyze and compare experimental results.
- Apply RCT methodology to Computer Application research projects.

Result

The Randomized Controlled Trial was successfully conducted using the example of an AI-based learning platform. The experimental group achieved **16.67% higher academic performance** than the control group, demonstrating the effectiveness of the AI-based learning system. The practical helped in understanding the methodology, applications, advantages, and significance of Randomized Controlled Trials in **Computer Applications** and research.

Practical No. 9

Title

Illustrate Various Steps of Literature Research: Narrative Review, Systematic Review, and Meta-analysis and Identify the Differences Among Them

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Aim

To study the various methods of literature research, namely Narrative Review, Systematic Review, and Meta-analysis, and identify the differences among them using a research topic in Computer Applications.

Objectives

1. To understand the concept of literature review.
2. To study the various steps involved in conducting literature research.
3. To understand Narrative Review, Systematic Review, and Meta-analysis.
4. To compare the three review methods.
5. To perform a literature review on a Computer Applications research topic.

Requirements

Hardware

- Computer/Laptop
- Internet Connection

Software

- Microsoft Word / Google Docs
- Google Chrome / Microsoft Edge

Resources

- Google Scholar
- IEEE Xplore
- ACM Digital Library
- ResearchGate
- Scopus
- Research Methodology Textbook

Practical Activity

Step 1: Select a Research Topic

Research Topic

Cybersecurity Using Machine Learning

Step 2: Search Research Papers

Search at least **five research papers** using research databases.

Selected Research Papers

1. Machine Learning Techniques for Cybersecurity
2. Intrusion Detection Using Artificial Intelligence
3. Deep Learning for Malware Detection
4. Network Intrusion Detection Systems Using Machine Learning
5. AI-Based Cyber Threat Detection

Step 3: Prepare a Narrative Review

Read all five research papers and prepare a descriptive summary without following any predefined research protocol.

Focus on:

- Objectives
- Methodology
- Findings
- Advantages
- Limitations

Step 4: Prepare a Systematic Review

Conduct a structured literature review using the following process.

Research Question

How does Machine Learning improve Cybersecurity systems?

Search Databases

- Google Scholar
- IEEE Xplore
- ACM Digital Library

Inclusion Criteria

- Published after 2020
- Peer-reviewed journals
- English language
- Research related to Machine Learning in Cybersecurity

Exclusion Criteria

- Duplicate papers
- Conference abstracts only
- Non-English publications

Data Extraction

Collect information regarding:

- Machine Learning algorithm used
- Dataset
- Accuracy
- Security threats detected
- Limitations

Step 5: Prepare a Meta-analysis

Compare quantitative results reported by different research papers.

Example:

Study	Detection Accuracy
Study 1	95%
Study 2	97%
Study 3	94%
Study 4	96%
Study 5	98%

Average Accuracy

\[\frac{95+97+94+96+98}{5}=96\% \]

The selected Machine Learning models achieved an **average intrusion detection accuracy of 96%**.

Step 6: Differentiate Narrative Review, Systematic Review and Meta-analysis

Parameter	Narrative Review	Systematic Review	Meta-analysis
Purpose	General overview	Structured review	Statistical analysis
Search Strategy	Flexible	Predefined	Uses systematic review data
Bias	High	Low	Very Low
Data	Qualitative	Qualitative & Quantitative	Quantitative
Reliability	Moderate	High	Very High
Time Required	Less	Moderate	High

Step 7: Prepare a Short Report

Title

Literature Review on Cybersecurity Using Machine Learning through Narrative Review, Systematic Review and Meta-analysis

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Introduction

Cybersecurity has become one of the most critical areas in Computer Applications due to the rapid increase in cyber threats such as malware, phishing attacks, ransomware, and network intrusions. Traditional security systems often fail to detect newly emerging attacks because they rely on predefined rules and signatures. Machine Learning (ML) provides intelligent techniques that enable security systems to learn from historical data, identify suspicious activities, and detect unknown cyber threats automatically.

Before developing any Machine Learning-based cybersecurity solution, researchers perform a **literature review** to understand previous work, identify research gaps, compare different techniques, and improve future research. Literature reviews can be conducted using **Narrative Review**, **Systematic Review**, and **Meta-analysis**, each serving a unique purpose in research methodology.

Narrative Review

A Narrative Review summarizes previously published studies without following a strict research protocol. It provides a broad understanding of the selected research topic.

Example

Five research papers related to **Machine Learning in Cybersecurity** were reviewed. Most studies focused on intrusion detection systems, malware detection, phishing attack prevention, and anomaly detection. Researchers concluded that Machine Learning algorithms such as Decision Trees, Random Forest, Support Vector Machine (SVM), and Neural Networks significantly improve the accuracy of cyber threat detection while reducing false alarms.

Advantages

- Easy to conduct.
- Provides a broad understanding of the research topic.
- Useful for introductory research.

Limitations

- Subjective interpretation.
- No predefined methodology.
- Higher possibility of researcher bias.

Systematic Review

A Systematic Review follows a structured methodology to identify, evaluate, and summarize research studies.

Research Question

How effective are Machine Learning techniques in improving Cybersecurity systems?

Search Strategy

Research papers were collected from Google Scholar, IEEE Xplore, ACM Digital Library, and ResearchGate.

Inclusion Criteria

- Published between 2020 and 2025.
- Peer-reviewed journals.
- Research focused on Machine Learning applications in Cybersecurity.
- English language publications.

Exclusion Criteria

- Duplicate studies.
- Non-English articles.

- Conference abstracts without full papers.

Findings

The review found that Machine Learning significantly improves malware detection, intrusion detection, phishing detection, and network traffic analysis. Algorithms such as Random Forest, Support Vector Machine, and Deep Neural Networks achieved high detection accuracy while reducing manual effort.

Advantages

- Transparent methodology.
- Reliable findings.
- Reduces research bias.

Limitations

- Time-consuming.
- Requires careful planning.
- Needs well-defined research criteria.

Meta-analysis

Meta-analysis combines numerical results from multiple studies to produce a single statistical conclusion.

Example

Five studies reported the following cybersecurity detection accuracies.

Study	Detection Accuracy
Study 1	95%
Study 2	97%
Study 3	94%
Study 4	96%
Study 5	98%

Combined Result

Average Detection Accuracy

96%

This statistical result indicates that Machine Learning techniques provide highly accurate cybersecurity solutions across multiple research studies.

Advantages

- Produces statistically reliable conclusions.
- Combines evidence from multiple studies.
- Reduces uncertainty in research findings.

Limitations

- Requires quantitative data.
- Complex statistical calculations.
- Depends on the quality of selected studies.

Comparison of Literature Review Methods

Feature	Narrative Review	Systematic Review	Meta-analysis
Purpose	General understanding	Evidence-based review	Statistical conclusion
Research Method	Flexible	Structured	Statistical
Bias	High	Low	Very Low
Data Used	Qualitative	Qualitative & Quantitative	Quantitative
Reliability	Moderate	High	Very High
Complexity	Low	Medium	High

Applications in Computer Applications

Narrative Review

- Artificial Intelligence
- Cybersecurity
- Cloud Computing
- Software Engineering

Systematic Review

- Machine Learning
- Blockchain Technology
- Internet of Things (IoT)
- Data Science

Meta-analysis

- AI Model Accuracy
- Malware Detection Systems

- Network Security Performance
- Intrusion Detection Systems

Conclusion

Literature research plays a vital role in identifying existing knowledge, understanding current technologies, and discovering research gaps before beginning any new project. **Narrative Review** provides a broad overview of previous studies, **Systematic Review** offers structured and evidence-based findings, while **Meta-analysis** combines statistical data from multiple studies to generate highly reliable conclusions.

For Computer Application students, these review methods are valuable in research areas such as Cybersecurity, Artificial Intelligence, Machine Learning, Cloud Computing, and Data Science. Conducting a proper literature review improves research quality, supports informed decision-making, and contributes to the development of innovative software and intelligent systems.

Expected Output

After completing this practical, the student should be able to:

- Explain the concept of literature research.
- Conduct a Narrative Review.
- Perform a Systematic Review.
- Understand the basics of Meta-analysis.
- Compare the three literature review methods.
- Prepare a structured literature review report.

Result

The literature review process was successfully conducted using the research topic **"Cybersecurity Using Machine Learning."** Narrative Review, Systematic Review, and Meta-analysis were performed to study existing research, compare methodologies, and analyze quantitative results. The practical demonstrated that **Systematic Review and Meta-analysis provide more reliable and evidence-based conclusions than Narrative Review**, making them highly useful for research in **Computer Applications**, particularly in Cybersecurity and Machine Learning.

Practical No. 10

Title

Illustrate Literature Review under Peer Learning

Aim

To understand the concept of literature review under peer learning and perform a collaborative literature review on a Computer Applications research topic.

Objectives

1. To understand the concept of literature review.
2. To understand the importance of peer learning in research.
3. To conduct a collaborative literature review using research papers.
4. To identify research gaps and summarize findings.
5. To improve teamwork, communication, and analytical skills through peer learning.

Requirements

Hardware

- Computer/Laptop
- Internet Connection

Software

- Microsoft Word / Google Docs
- Google Chrome / Microsoft Edge
- Microsoft PowerPoint (Optional)

Resources

- Google Scholar
- IEEE Xplore
- ACM Digital Library
- ResearchGate
- Scopus
- Research Methodology Textbook

Practical Activity

Step 1: Select a Research Topic

Research Topic

Artificial Intelligence in Smart Agriculture

Step 2: Form a Peer Learning Group

Create a group of **4–5 students** and assign individual responsibilities.

Student	Responsibility
Student 1	Search research papers
Student 2	Read and summarize research papers
Student 3	Compare research findings
Student 4	Prepare the literature review report
Student 5	Review, edit and finalize the report

Step 3: Search Research Papers

Search at least **five peer-reviewed research papers** using:

- Google Scholar
- IEEE Xplore
- ACM Digital Library

Selected Research Papers

1. Artificial Intelligence in Smart Agriculture
2. Machine Learning for Crop Prediction
3. IoT-Based Smart Farming Systems
4. Deep Learning for Plant Disease Detection
5. AI-Powered Precision Agriculture

Step 4: Individual Literature Review

Each student reviews one research paper and records the following information:

- Research Objective
- Research Methodology
- Technologies Used
- Results
- Advantages

- Limitations

Step 5: Peer Discussion

Conduct a group discussion.

Discuss the following:

- Similarities among the papers
- Differences in methodologies
- Research gaps
- Future scope
- Suggestions for improvement

Record the common observations.

Step 6: Prepare a Collaborative Literature Review

Prepare one combined literature review including:

- Introduction
- Summary of each research paper
- Comparative analysis
- Research gaps
- Future scope
- Conclusion

Step 7: Prepare a Presentation (Optional)

Prepare a short PowerPoint presentation summarizing:

- Research Topic
- Literature Review
- Findings
- Research Gaps
- Conclusion

Present the findings before classmates.

Step 8: Prepare a Short Report

Title

Collaborative Literature Review on Artificial Intelligence in Smart Agriculture through Peer Learning

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Introduction

A **literature review** is a systematic process of collecting, studying, evaluating, and summarizing existing research related to a particular topic. It helps researchers understand previous work, identify research gaps, and build a strong foundation for future studies.

Peer learning is a collaborative educational approach in which students work together to learn from one another by sharing knowledge, discussing ideas, reviewing research papers, and solving problems collectively. In research methodology, peer learning enhances communication, teamwork, critical thinking, and analytical skills.

For this practical, a peer learning approach was used to conduct a literature review on **Artificial Intelligence in Smart Agriculture**, allowing each student to contribute equally to the research process.

Literature Review

Five research papers related to **Artificial Intelligence in Smart Agriculture** were selected and reviewed collaboratively.

Paper 1: Artificial Intelligence in Smart Agriculture

This study explained how Artificial Intelligence helps farmers monitor crop growth, detect diseases, and improve agricultural productivity using intelligent decision-making systems.

Paper 2: Machine Learning for Crop Prediction

The paper focused on Machine Learning algorithms that analyze weather conditions, soil quality, rainfall, and historical crop data to predict crop yield accurately.

Paper 3: IoT-Based Smart Farming Systems

This research discussed Internet of Things (IoT) devices that collect real-time environmental data using sensors. AI algorithms analyze this data to automate irrigation, fertilization, and crop monitoring.

Paper 4: Deep Learning for Plant Disease Detection

This paper presented Deep Learning techniques that identify plant diseases from leaf images. The proposed system improves disease detection accuracy and helps farmers take preventive actions.

Paper 5: AI-Powered Precision Agriculture

This study described precision farming techniques using drones, satellite imagery, GPS technology, and Artificial Intelligence for efficient resource utilization and sustainable farming.

Peer Learning Discussion

After reviewing all research papers, the peer group discussed the following observations.

Similarities

- All studies focused on improving agricultural productivity.
- Artificial Intelligence and Machine Learning were the primary technologies.
- Data analysis was essential in all research papers.
- Automation was a common objective.

Differences

- Different Machine Learning algorithms were used.
- Some studies emphasized crop prediction, while others focused on disease detection and irrigation management.
- Various datasets and evaluation techniques were adopted.

Research Gaps

- Lack of large agricultural datasets.
- High implementation cost.
- Limited adoption among small-scale farmers.
- Need for multilingual AI applications for rural communities.

Future Scope

- AI-integrated agricultural robots.
- Drone-based crop monitoring.
- Smart irrigation systems.
- Predictive farming using Big Data.
- Sustainable agriculture through Artificial Intelligence.

Benefits of Peer Learning

- Improves teamwork.
- Encourages collaborative learning.
- Enhances communication skills.

- Develops critical thinking.
- Promotes knowledge sharing.
- Increases confidence in research activities.
- Reduces individual workload.

Applications in Computer Applications

Peer learning literature reviews are widely used in:

- Artificial Intelligence
- Machine Learning
- Data Science
- Cybersecurity
- Software Engineering
- Cloud Computing
- Internet of Things (IoT)
- Blockchain Technology
- Mobile Application Development

Comparison Between Individual Learning and Peer Learning

Parameter	Individual Learning	Peer Learning
Learning Style	Independent	Collaborative
Knowledge Sharing	Limited	High
Communication	Low	High
Problem Solving	Individual	Group Discussion
Critical Thinking	Moderate	High
Research Quality	Moderate	Better
Time Required	More	Less

Conclusion

Peer learning is an effective approach to conducting literature reviews because it encourages collaboration, discussion, and knowledge sharing among students. By reviewing different research papers and discussing their findings collectively, students gain a deeper understanding of the research topic and improve their analytical abilities.

In this practical, the collaborative literature review on **Artificial Intelligence in Smart Agriculture** demonstrated how peer learning enhances research quality by combining the strengths of multiple learners. It also helped identify current research trends, existing challenges, and future opportunities in Artificial Intelligence applications for agriculture.

For Computer Application students, peer learning strengthens research methodology skills and prepares them for collaborative software development, project work, and industrial research environments.

Expected Output

After completing this practical, the student should be able to:

- Explain the concept of literature review.
- Understand peer learning in research.
- Conduct a collaborative literature review.
- Analyze and compare multiple research papers.
- Identify research gaps and future research opportunities.
- Prepare a professional literature review report through teamwork.

Result

The literature review under **peer learning** was successfully completed using the topic **"Artificial Intelligence in Smart Agriculture."** Each group member contributed by reviewing different research papers, discussing the findings, and preparing a collaborative report. The practical demonstrated that peer learning improves research quality, enhances teamwork, develops communication skills, and enables students to gain a broader understanding of current research trends in **Computer Applications**.
